TEXORA · LECTURE SHEET

Polymer Structure and Classifications

What a polymer is, how its chains are built and arranged, and the seven independent ways of classifying polymers — the vocabulary on which the whole of fibre science rests.

TE-2107 · POLYMER SCIENCE AND ENGINEERING · TOPIC 1 · LEVEL 2 TERM 1

1. Basic definitions

A **polymer** is a very large molecule (a **macromolecule**) built by joining many small molecules, called **monomers**, into a long chain through covalent bonds. The chemical process of joining them is **polymerisation**. The small repeating block inside the chain is the **repeat unit**; for many polymers the repeat unit is identical with the monomer, but in condensation polymers it differs from the monomers by the small molecule (usually water) that is eliminated.

The number of repeat units in one chain is the **degree of polymerisation (DP)**. The molar mass of the polymer molecule follows directly:

$$M = DP \times M_0$$

where M_0 is the molar mass of the repeat unit. Short chains of only a few units (DP roughly 2–10) are called **oligomers**; they are liquids or soft solids, not useful materials. Fibre-forming polymers typically have DP values from about 100 (nylon, polyester) to several thousand (polyethylene, cellulose in cotton has DP of several thousand). Real samples are never uniform: every polymer sample contains chains of many different lengths, so its molar mass is an *average* — a fact developed fully in Topic 4.

WORKED EXAMPLE — DP FROM MOLAR MASS

A nylon 6,6 sample has a number-average molar mass of 22,632 g/mol. The repeat unit, formed from hexamethylenediamine (116.21 g/mol) and adipic acid (146.14 g/mol) with the loss of two water molecules (2 × 18.02), is $M_0 = 116.21 + 146.14 - 36.03 = 226.32$ g/mol.

$DP = M \div M_0 = 22{,}632 \div 226.32 =$ **100**. Each average chain therefore contains 100 repeat units — 100 diamine and 100 diacid residues. The same arithmetic for a polyester of M = 19,217 g/mol with $M_0 = 192.17$ g/mol (Topic 3) likewise gives DP = 100.

2. Anatomy of a polymer chain

2.1 Homopolymers and copolymers

A chain built from **one** kind of monomer is a **homopolymer**: polyethylene from ethylene, nylon 6 from caprolactam. A chain built from **two or more** monomers is a **copolymer**, and the monomers may be arranged in four distinct architectures:

- **Random copolymer** — the two units (A and B) are scattered irregularly: –A–B–A–A–B–B–A–B–. Acrylic fibre is usually a random copolymer of acrylonitrile with a few percent of a comonomer added to accept dye.
- **Alternating copolymer** — strict alternation: –A–B–A–B–A–B–.
- **Block copolymer** — long runs of one monomer joined to long runs of the other: –A–A–A–A–B–B–B–B–. Spandex (elastane) is a block copolymer: soft polyether or polyester blocks give stretch, hard urethane blocks act as anchor points.
- **Graft copolymer** — a backbone of one polymer carrying side branches of another, like a tree trunk with limbs.

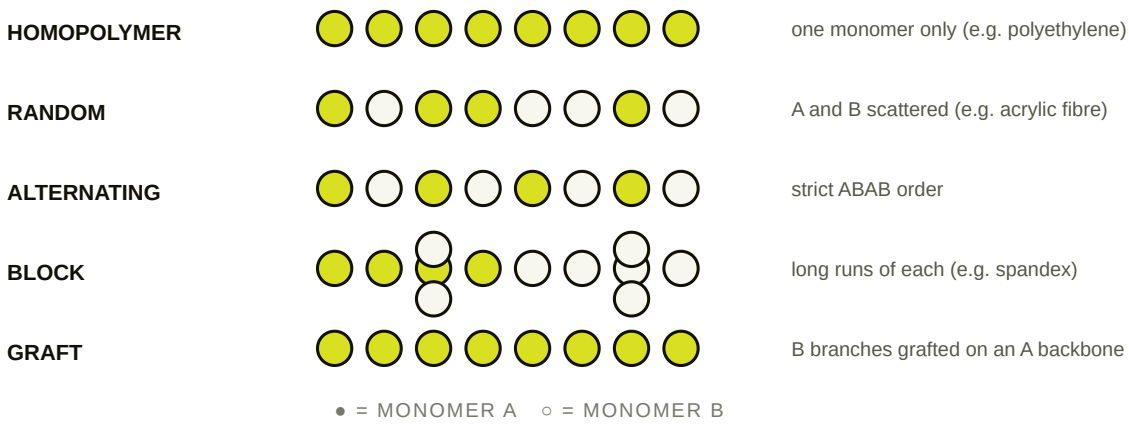


Figure 1 — Chain architectures. The same two monomers give four different copolymers depending only on how the units are sequenced — and each sequence gives different properties.

2.2 Linear, branched and cross-linked chains

- **Linear polymers** — the units form one long unbranched chain (HDPE, nylon, polyester). Chains can slide past one another when heated, so linear polymers can be melted and reshaped.
- **Branched polymers** — side chains grow from the main backbone (LDPE). Branches prevent close packing, so branched polymers are less crystalline, softer and lower-melting than their linear versions.
- **Cross-linked (network) polymers** — neighbouring chains are joined by covalent bridges into one giant three-dimensional network (vulcanised rubber, thermoset resins, wool's disulfide bonds partly behave this way). A network cannot melt or dissolve; it only swells in solvents.

2.3 Stereochemistry: tacticity

Where a monomer carries a side group (e.g. the $–CH_3$ of polypropylene), the side groups may be arranged **isotactically** (all on the same side — regular, crystallisable), **syndiotactically** (alternating sides — also regular) or **atactically** (random sides — irregular, cannot crystallise, soft and amorphous). Only isotactic polypropylene makes a useful fibre; atactic polypropylene is a sticky waste product of early production. Regularity of structure is the first condition of crystallinity, and crystallinity governs strength.

3. Bonds within and between chains

Atoms *within* a chain are held by strong **covalent bonds** — breaking the chain itself needs high energy, which is why polymers are strong along the fibre axis. Chains are held *to each other* only by much weaker **secondary forces**:

- **Van der Waals forces** — weak attractions between all molecules; the only cohesion in polyethylene.
- **Dipole–dipole forces** — between polar groups, e.g. the nitrile (–C≡N) groups of acrylic.
- **Hydrogen bonds** — the strongest secondary force, between N–H and C=O groups; they link nylon chains into sheets and hold cellulose chains together, giving both polymers their high strength and melting behaviour.

Drawing a fibre aligns the chains, multiplies these contacts, and lets crystallites form — the structural reason drawn fibres are several times stronger than the same polymer undrawn.

4. Classifications of polymers

Polymers are classified independently on several bases; a single polymer sits in one class of *each* scheme.

4.1 By origin (source)

CLASS	MEANING	EXAMPLES
Natural	Occur ready-made in nature	Cotton and flax cellulose, wool keratin, silk fibroin, natural rubber
Semi-synthetic	Natural polymer chemically modified	Viscose and lyocell (from wood cellulose), cellulose acetate
Synthetic	Built from small petrochemical monomers	Polyester, nylon, acrylic, polypropylene

4.2 By polymerisation reaction

- **Addition (chain-growth) polymers** — monomers add one by one to a growing chain, usually across a C=C double bond; no by-product. Examples: polyethylene, polypropylene, PVC, acrylic.
- **Condensation (step-growth) polymers** — monomers react through functional groups (–OH, –COOH, $–NH_2$), eliminating a small molecule (water, methanol) at each step. Examples: polyester (PET), nylon 6,6.

4.3 By response to heat

- **Thermoplastics** — soften and melt on heating, harden again on cooling, repeatedly (nylon, polyester, polypropylene). Melt spinning and heat-setting exist only because of this class.
- **Thermosets** — set permanently by cross-linking during curing; they char instead of melting (phenol-formaldehyde, melamine resins, vulcanised rubber).

4.4 By structure

Linear, branched and cross-linked, as in §2.2.

4.5 By crystallinity

Crystalline, semi-crystalline and amorphous. No flexible-chain polymer is 100 % crystalline; textile fibres are **semi-crystalline** — ordered crystalline regions embedded in disordered amorphous regions. The amorphous regions admit water and dye; the crystalline regions give strength and resistance to solvents. Amorphous polymers (atactic polystyrene, acrylic is nearly so) are glassy or rubbery and weak as fibres.

4.6 By end use / family

Plastics, fibres, rubbers (elastomers), coatings and adhesives — the same chemistry sorted by the form in which it is used. Fibre-forming polymers are a special subset: they must be linear, regular enough to crystallise on drawing, of adequate DP, and meltable or soluble for spinning.

4.7 By inter-chain forces (fibre science view)

Elastomers (weak forces, huge reversible extension), fibres (strong forces, high crystallinity — nylon, polyester) and plastics (intermediate) — a ranking by how much the secondary forces of §3 dominate behaviour.

Glossary

Monomer — the small molecule from which a polymer is built. **Repeat unit** — the smallest block that repeats along the chain. **Degree of polymerisation (DP)** — the number of repeat units per chain.

Oligomer — a chain of only a few units. **Homopolymer / copolymer** — chains from one / two-or-more monomer types. **Tacticity** — the regularity of side-group arrangement along the chain. **Cross-link** — a covalent bridge between two chains. **Semi-crystalline** — containing both ordered crystalline and disordered amorphous regions.

Key points

• $M = DP \times M_0$; fibre polymers need DP from about 100 upward, and every sample contains a spread of chain lengths.
• Copolymer architecture (random, alternating, block, graft) tunes properties — spandex's stretch comes from its block structure.
• Covalent bonds run along the chain; van der Waals, dipole and hydrogen bonds act between chains — drawing multiplies inter-chain contact, hence strength.
• Classify on every axis at once: e.g. PET is synthetic, condensation, thermoplastic, linear and semi-crystalline.
• Regular, linear chains crystallise; irregular or heavily branched chains stay amorphous and weak.

Practice

1. A PET sample has M = 28,826 g/mol and a repeat unit of 192.17 g/mol. Calculate its DP.
2. Classify each of the following on all applicable axes (origin, reaction, heat response, structure): (a) nylon 6,6; (b) viscose; (c) LDPE; (d) vulcanised rubber.
3. Why is atactic polypropylene useless as a fibre while isotactic polypropylene makes strong fibres?
4. Spandex stretches and recovers; ordinary polyester does not. Explain in terms of copolymer architecture.
5. A polymer sample is found to be insoluble in every solvent and does not melt, but swells in toluene. What structural class does it belong to, and why does it only swell?

Answer key

1. DP = 28,826 ÷ 192.17 = **150**.
2. (a) Synthetic, condensation, thermoplastic, linear. (b) Semi-synthetic (regenerated natural cellulose), thermoplastic in processing terms but degrades before melting — class it as a modified natural polymer, linear. (c) Synthetic, addition, thermoplastic, branched. (d) Natural (or synthetic) rubber made into a cross-linked network: thermoset behaviour after vulcanisation.
3. Atactic chains have randomly placed methyl groups, cannot pack into crystals, and stay soft and amorphous with weak inter-chain contact; isotactic chains are regular, crystallise on drawing, and the crystallites give strength and a useful melting point.

4. Spandex is a block copolymer: long soft blocks coil and uncoil for stretch, while glassy hard blocks cluster at intervals as physical cross-links that pull the chain back and prevent permanent flow. Polyester is a homopolymer of uniform stiff units — no soft blocks, no elastic recovery mechanism.
5. It is a cross-linked (network) polymer. Solvent molecules can penetrate and push the network outward (swelling), but the chains cannot separate and dissolve because covalent cross-links tie the whole sample into one molecule; for the same reason it cannot melt.

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